AUTONOMOUS ML RESEARCH / TECHJAM 2026

SciOdyssey: Long-Horizon Autonomous ML Research Agent

Autonomous by default. Supervisable by design.

ROBUST USER-FRIENDLY BROAD SEARCH

01 / PROBLEM · THE RESEARCH TASK

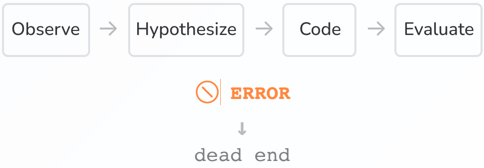
Can an agent carry a research task to completion?

Explore	Choose hypotheses, write code, and run experiments.	Scientific judgment
Recover	Handle failures and revise unsuccessful ideas.	Long-running work
Deliver	Retain a valid model and the evidence behind it.	Verifiable progress

01 / PROBLEM · FAILURE MODES

Why current agents fail

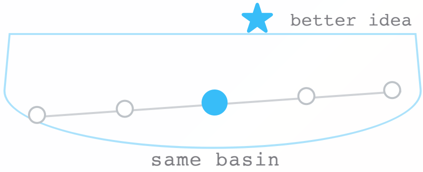
01 WORKFLOW



Fixed recovery paths

Unexpected errors need human help.

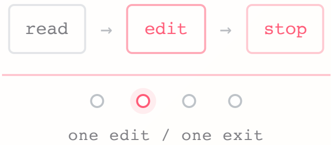
02 SINGLE AGENT



History shapes search

Errors can persist

03 TREE SEARCH



Repeated context reads

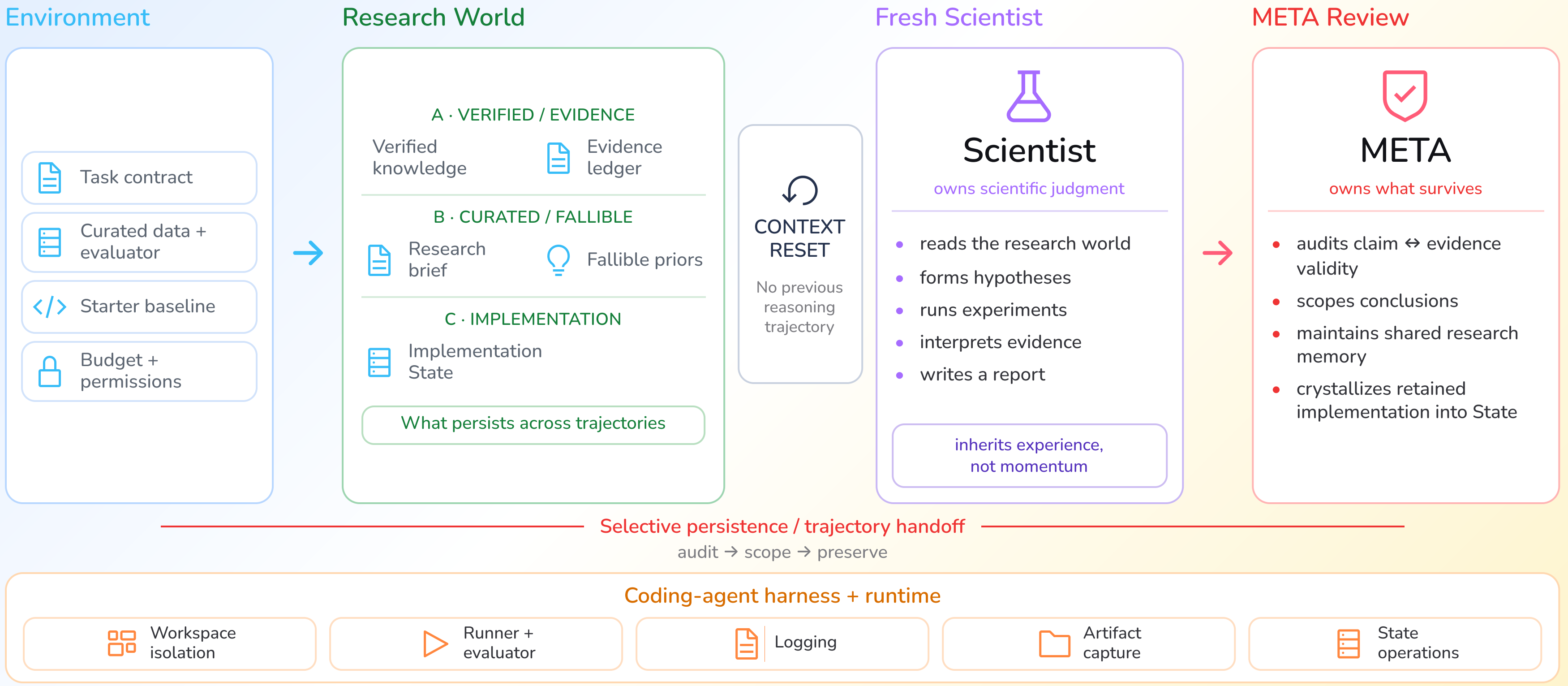
Each small edit pays the reading cost.

- Adaptive tools
- Fresh reasoning
- Audited evidence
- Reusable context

Build the system →

02 / SYSTEM · DESIGN

A system for sustained research



SciOdyssey

A research layer over your agent harness.

Your agent. One persistent research world.

04 / EVALUATION

CLAIM

Our agent can carry a research task to completion.

KUAIRAND-PURE • FIXED EVALUATOR • PUBLIC VALIDATION

01 MEASURED RESULT

The model got better.

02 SUSTAINED SEARCH

The search kept going.

03 RECOVERY UNDER FAILURE

Our agent is robust.

04 DIRECT CODEX CONTROL

Our agent outperforms Codex.

05 RESOURCE ACCOUNTING

We are not tokenmaxxing.

05 / INSIGHTS · WHAT THE RUN TAUGHT US

The run taught us more than the final score.

One durable lesson from the experiments.

01

Diversity is a prior.

Keep `gemini-3.7-flash` as META while Scientist rotates across `gpt-5.6-sol`, `gemini-3.7-flash`, and `gpt-5.6-luna`. This widens priors; it does not prove the gain.

A clean repo. A continuous agent.

FIELD NOTE / 01

Engineering practices to apply next: isolate parallel changes and make unattended work reviewable.

01 MANY AGENTS

Give every agent a branch.

main
clean trunk

- agent A / branch
- agent B / branch
- agent C / branch

review
→ merge

Let agents work independently. Conflicts stay local, and the main branch stays clean.

02 NO QUOTA / NO LAPTOP

Leave the next move in GitHub.

01 Issue / prompt — inject the task
→
02 GitHub Action — run the agent
→
03 Commit — review the change

Connect the agent to GitHub Actions. It can code and commit while you are offline; review before merging.

PROPOSED PRACTICE Workflow guidance; GitHub continuation is not a benchmarked system feature.

Keep the experience.
Reopen the search.

```
# From the configured repo, run one autonomous cycle
$ ./scripts/research-agent step --cli codex \
    --target /absolute/path/to/project --allow-edits
```

github.com/zc6600/research-agent-kuairand ↗

Code · Technical report · Experiment ledger · Checked output

APPENDIX / OPEN-WORLD RESEARCH

Research is an open-world task. We need an open-world coding agent.

01 OPEN ACTION SPACE

- Shell
- Files
- Browser
- Git

Use what exists

02 SELF-RECOVERY

- Read errors
- Inspect docs
- Inspect source
- Install packages

Fix what breaks

03 RUNTIME EXTENSION

- MCP
- Skills
- CLI
- Packages

Add what's missing

WORKFLOW EXAMPLE • LANGGRAPH

LangGraph makes a known workflow explicit as nodes + edges.

Missing capability? Leave the graph → inspect → extend → retry

APPENDIX / THE TEAM

Five people. One autonomous research loop.

OVERLAPPING RESPONSIBILITIES / SHARED WORLD

CZ

Chen Zhu

TEAM LEAD / SYSTEM

Direction · architecture · integration

ZZ

Zhou Ziyu

MODELING / EXPERIMENTS

Features · ensembles · evidence review

SX

Shilin Xu

DATA / METRICS

Data workflow · contracts · alignment

GG

GE GAO

RUNTIME / RELIABILITY

Execution · testing · integration support

JL

Jiran Li

EVIDENCE / REPRODUCIBILITY

Telemetry · documentation · packaging

SCIODYSSEY TEAM

Every handoff kept the research world moving.

Research is an open-world task.
We need an open-world coding agent.

LANGGRAPH-STYLE ORCHESTRATION

predefined graph

fixed tool set

↓

Tool A | Tool B | Tool C

API fails

↓

no recovery tool

↓

DEAD END

Capabilities are bounded by what developers anticipated.

CODING-AGENT HARNESS

open action space

Computer

↓

shell | files | browser

↓

git | docs | packages

skills → agent ← MCP

↓

CLI / API

fail → inspect → learn → modify environment → retry

The agent can acquire the missing capability at runtime.